

6E Interpolation and extrapolation

The equation of the line of best fit or regression line can provide important information and be used to make predictions. The gradient (m) indicates the change in dependent variable as the independent variable increases by 1 unit. The vertical intercept (b) indicates the value of the dependent variable when the independent variable is zero. In addition to this information, the equation of best fit is used for interpolation and extrapolation.

Interpolation

Interpolation is the use of the linear regression line to predict values within the range of the dataset. If the data has a strong linear association then we can be confident our predictions are accurate. However, if the data has a weak linear association, we are less confident with our predictions.

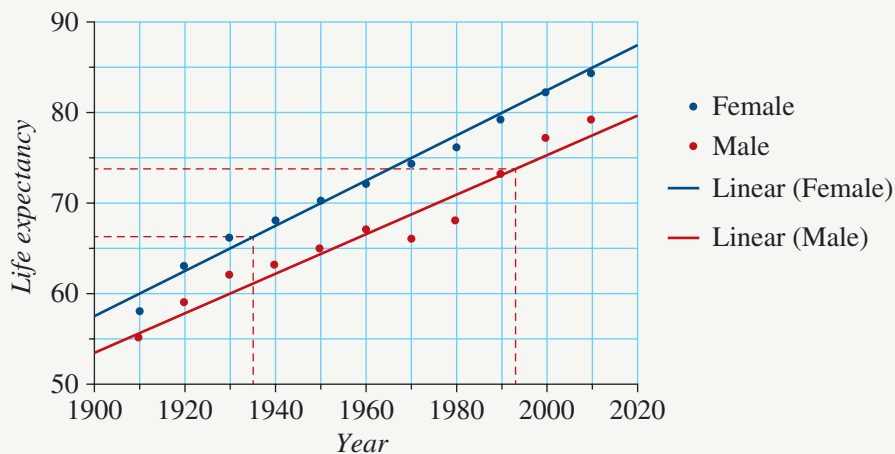


Example 7: Making predictions using interpolation

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Life expectancy at birth for females and males is shown below.

Year	1910	1920	1930	1940	1950	1960	1970	1980	1990	2000	2010
Female	58	63	66	68	70	72	74	76	79	82	84
Male	55	59	62	63	65	67	66	68	73	77	79



- What was the life expectancy in 1935 for females?
- What was the life expectancy in 1995 for males?

SOLUTION:

- Draw a vertical line from 1935 until it intersects the blue line. At this point draw a horizontal line until it reaches the vertical axis. Read the value.
 - Draw a vertical line from 1995 until it intersects the red line. At this point draw a horizontal line until it reaches the vertical axis. Read the value.
- Life expectancy for females in 1935 is approximately 67 years.
 - Life expectancy for males in 1995 is approximately 74 years.



Extrapolation

Extrapolation is the use of the linear regression line to predict values outside the range of the dataset. Predicted values are either smaller or larger than the dataset. The accuracy of predictions using extrapolation depends on the strength of the linear association similar to interpolation. It may not be reasonable to extrapolate too far as this example shows.



Example 8: Making predications using extrapolation

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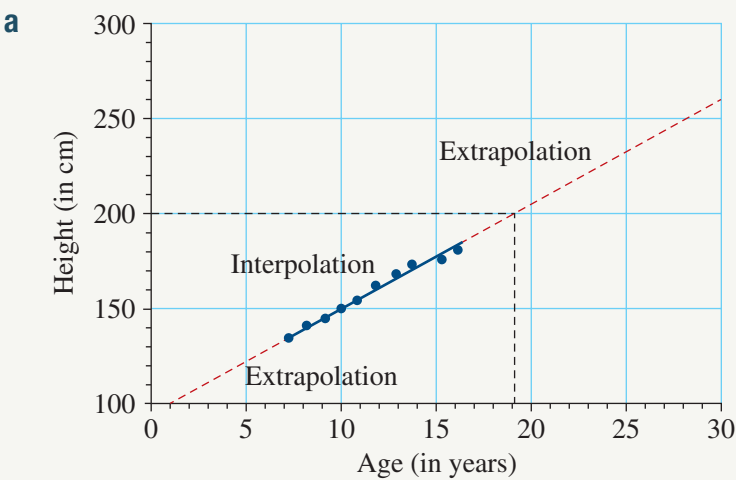
The table below shows the age of a student and their height in centimetres.

Age (in years)	7	8	9	10	11	12	13	14	15	16
Height (in cm)	133	139	144	149	156	163	170	174	177	181

- a Construct a scatterplot from the table using age from 0 to 30 and height from 100 to 300.
- b Draw the line of best fit and describe the association between age and height.
- c Predict the height of the student when they are aged 19 years.
- d What are the limitations of this linear model?

SOLUTION:

- 1 Draw a number plane with age as the horizontal axis and height as the vertical axis.
- 2 Determine a scale for the horizontal axis. Let each unit represent 1 year.
- 3 Determine a scale for the vertical axis. Let each unit represent 10 cm.
- 4 Write a title for the horizontal and vertical axes.
- 5 Plot the points (7, 133) (8, 139) (9, 144),...
- 6 There is a small amount of scatter in the scatterplot.
- 7 Read the height from the scatterplot when age is 19.
- 8 Extrapolation too far from the dataset needs to be done carefully.



- a Strong positive linear association.
- b Height of the student is 200 cm when they are 19 years old.
- c Adult height does not grow at the same rate as a child. Using the model to extrapolate is flawed, e.g., the prediction is the height will be 260 cm at age 30.

INTERPOLATION

EXTRAPOLATION

Predicting values within the dataset range

Predicting values outside the dataset range

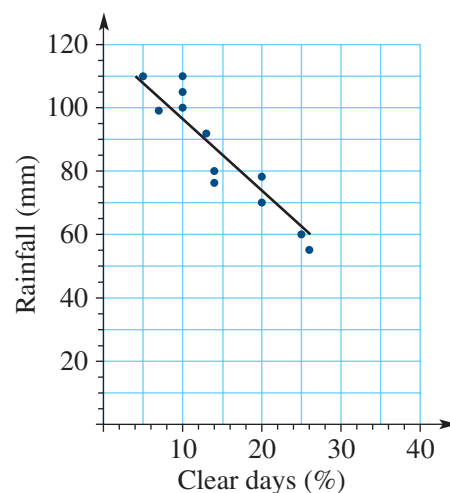
Exercise 6E

LEVEL 1

Example 7

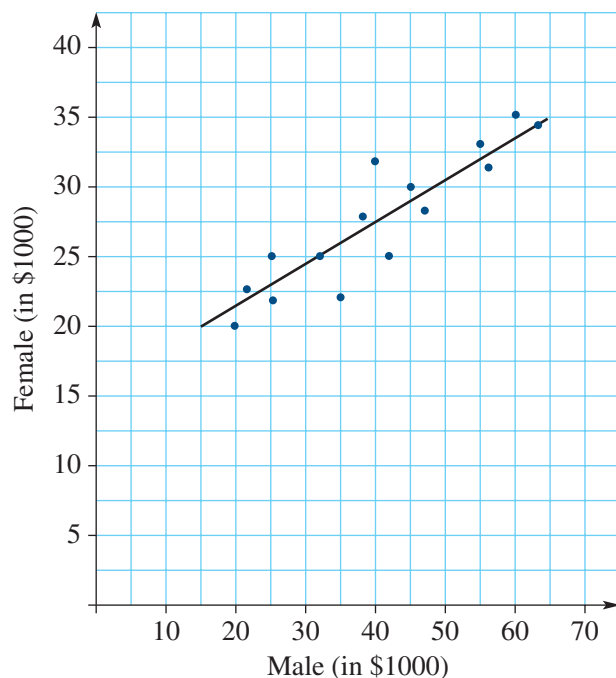
- 1 The scatterplot opposite shows the rainfall (in mm) and the percentage of clear days for each month in 2018.

- How many months had 10% of clear days?
- What was the percentage of clear days when the rainfall was 70 mm?
- Predict the rainfall in the month given the following percentage of clear days:
 - 4%
 - 22%
 - 26%.
- Predict the percentage of clear days in the month given the following rainfall:
 - 80 mm
 - 90 mm
 - 100 mm.



- 2 The scatterplot shows the average annual female income plotted against average annual male income for 15 countries.

- What was the average annual female income for a country whose average annual male income was \$45 000?
- How many countries had an average annual female income of \$25 000?
- Predict the female income given the following male income:
 - \$20 000
 - \$40 000
 - \$60 000.
- Predict the male income given the following female income:
 - \$25 000
 - \$30 000
 - \$35 000.



- 3 In a study of errors made in a test, for times ranging from 5 to 25 seconds, the equation relating the number of errors to time is:

$$\text{errors} = -0.53 \times \text{time} + 15$$

Use this equation to predict the number of errors (to nearest whole number) with the following times. Are you interpolating or extrapolating?

- 10 seconds
- 20 seconds
- 30 seconds